

Performance Testing

Load, stress, spikes, endurance, scalability, metrics and tools such as JMeter and k6.

What Performance Testing measures

Performance Testing evaluates response time, throughput, resource usage, stability and capacity under different load conditions.

Load Testing

Load Testing verifies the system under expected or planned load. The goal is not merely to generate users, but to determine whether performance objectives are met.

Stress Testing

Stress Testing pushes the system beyond normal conditions to observe degradation, limits, failures and recovery capability.

Spike, Soak and Scalability

Spike evaluates sudden changes; Soak/Endurance maintains load for a long period; Scalability evaluates how the system responds to growing demand and resources.

Important metrics

Response time, percentiles such as p95/p99, throughput, error rate, CPU, memory, connections and queues help explain behavior. Averages alone can hide poor experience for some users.

JMeter, k6 and Gatling

JMeter provides a mature UI and broad ecosystem; k6 favors tests as code; Gatling also supports programmable load scenarios. Tool choice depends on context.

Test preparation

Define objectives, load, data, environment, duration, ramp-up, success criteria and monitoring before execution. An unrealistic load model can produce misleading conclusions.

Analysis and diagnosis

When degradation occurs, correlate load results with logs, metrics, traces and infrastructure. Performance is a system concern, not just a test-tool concern.

QA tip

Define success criteria before the test; generating heavy load without an objective does not produce a useful conclusion.

Official resources

Site oficial: <https://jmeter.apache.org/>